

Happy Supplies Parts Warehouse

customer_and_parts

<u>customerNumber</u> PK FK	<u>partNumber</u> PK FK	<u>date</u> PK	time	employee	qtyOrdered
HG54587	10654	7/1/2024	10:30am	D. Harrison	12
HG54587	10456	7/1/2024	10:30am	D. Harrison	7
HG54587	10776	7/1/2024	10:30am	D. Harrison	9
HG54587	10657	7/1/2024	10:30am	D. Harrison	10

<u>customerNumber</u> PK	customerName	customerType
HG54587	Jeff Peterson	Consumer

<u>partNumber</u> PK	name	type	cageCode	unitPrice
10654	Float Control	Plumbing	G413	12
10456	Modulator	Electrical	H433	7
10776	Hose Assembly	Plumbing	G413	9
10657	Float Assembly	Plumbing	G413	10

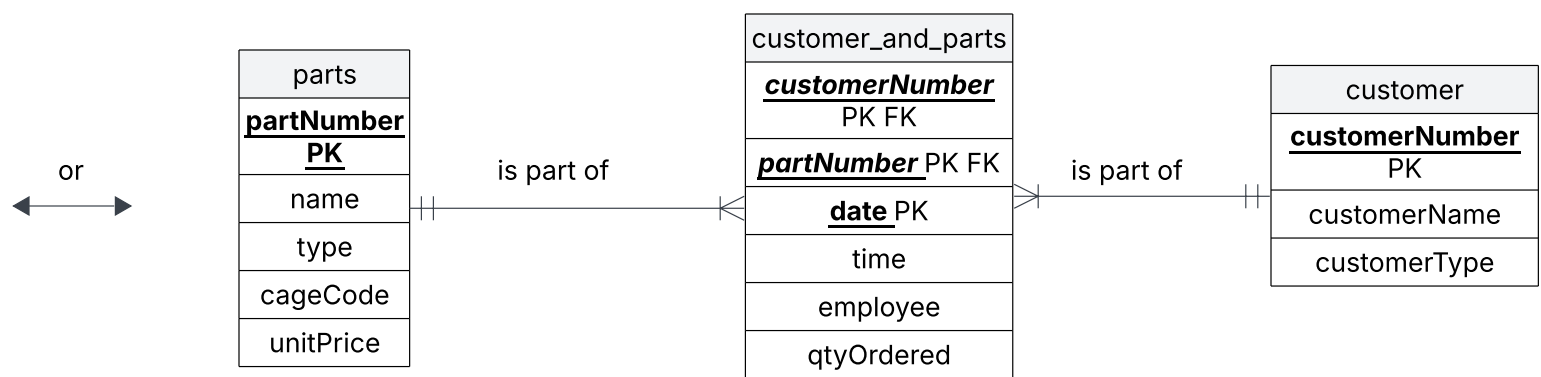
original_Table

<u>customerNumber</u> PK	customerName	<u>partNumber</u> PK	customerType	<u>date</u> PK	time	employee	name	type	cageCode	qtyOrdered	unitPrice
HG54587	Jeff Peterson	10654	Consumer	7/1/2024	10:30am	D. Harrison	Float Control	Plumbing	G413	4	12
HG54587	Jeff Peterson	10456	Consumer	7/1/2024	10:30am	D. Harrison	Modulator	Electrical	H433	3	7
HG54587	Jeff Peterson	10776	Consumer	7/1/2024	10:30am	D. Harrison	Hose Assembly	Plumbing	G413	7	9
HG54587	Jeff Peterson	10657	Consumer	7/1/2024	10:30am	D. Harrison	Float Assembly	Plumbing	G413	5	10

- Assumptions:
- customerNumber is unique and links back to only one person
 - partNumber can be repeated (different people can buy the same product)

- Comments:
- The PK for this table is a combination of customerNumber, partNumber, and date. This is because if date wasn't part of the composite key, then the same customer can order the same item, but on a different day, which would result in two rows having the same PK values and we would not be able to distinguish between the two of them
 - Currently, the table is in 1NF

Normalizing to 2NF/3NF; creating tables for each of the partial dependencies for customerNumber and partNumber. Once it is in 2NF, no transitive dependencies exist, so the table is normalized to 3NF



Panacea Mental Health Corporatoin

<u>staffNo</u> PK	<u>date</u> PK	<u>time</u> PK	patNo	patName
S1011	9/12/2022	10:00	P100	Lily White
S1011	9/12/2022	12:00	P105	Jill Baker
S1024	9/12/2022	10:00	P108	Andy McKee
S1024	9/14/2022	14:00	P108	Andy McKee
S1032	9/14/2022	16:30	P105	Jill Baker
S1032	9/15/2022	18:00	P110	Jimmy Winter

<u>staffNo</u> PK	therapistName
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard levin

<u>staffNo</u> PK FK	<u>branchID</u> PK FK
S1011	M15
S1024	Q10
S1032	M15
S1032	B13

<u>branchID</u> PK	branch
1	M15
2	Q10
3	B13

Normalizing to 3NF by creating a table for all transitive dependencies, which applies to the patNo and patName fields

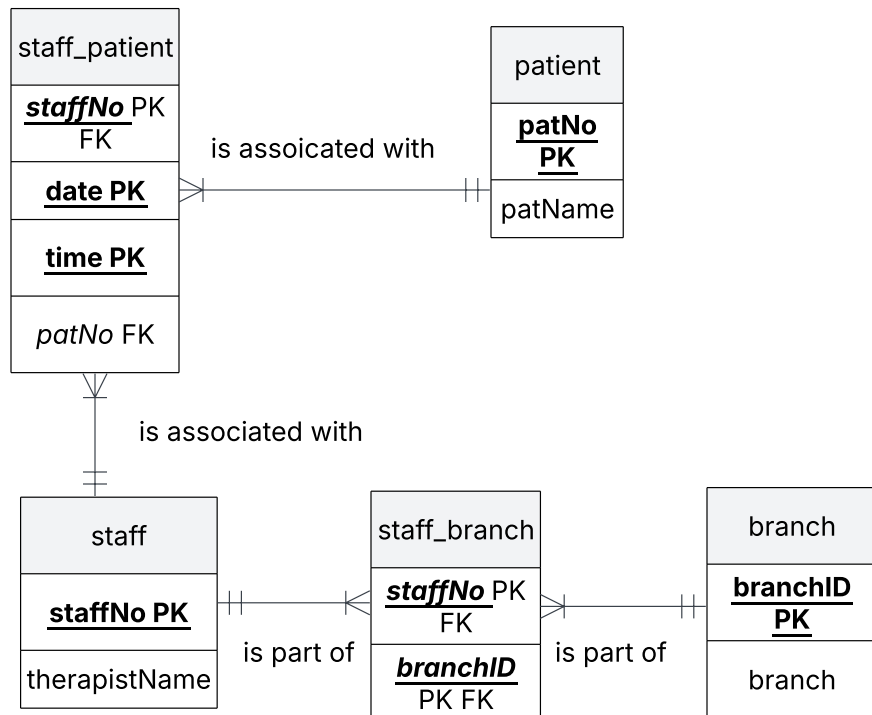
<u>staffNo</u> PK FK	<u>date</u> PK	<u>time</u> PK	patNo FK
S1011	9/12/2022	10:00	P100
S1011	9/12/2022	12:00	P105
S1024	9/12/2022	10:00	P108
S1024	9/14/2022	14:00	P108
S1032	9/14/2022	16:30	P105
S1032	9/15/2022	18:00	P110

<u>patNo</u> PK	patName
P100	Lily White
P105	Jill Baker
P108	Andy McKee
P110	Jimmy Winter

<u>staffNo</u> PK	therapistName
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard levin

<u>staffNo</u> PK FK	<u>branchID</u> PK FK
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard levin

<u>branchID</u> PK	branch
1	M15
2	Q10
3	B13



The Maid Better Temp Agency

original_table

<u>eNo</u>	<u>contractNo</u>	hours	eName	eventNo	eventLoc
1135	C1024	16	Smith J	H25	Queens
1057	C1024	24	Hocine D	H25	Queens
1068	C1025	28	White T	H4	Yonkers
1135	C1025	15	Smith J	H4	Yonkers
1135	C1026	10	Smith J	H25	Queens

- Comments:
- Since no single field is unique, having a combination of two fields, such as eNo and contractNo, will work as a PK
 - Currently, the table is in 1NF

Normalizing to 2NF by creating separate tables for partial dependencies for the eNo and contractNo fields

<u>eNo</u>	<u>contractNo</u>	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10

<u>eNo</u>	eName
1057	Hocine D
1068	White J
1135	Smith J

<u>contractNo</u>	eventNo	eventLoc
C1024	H25	Queens
C1025	H4	Yonkers
C1026	H25	Queens

Normalizing to 3NF by creating tables for any transitive dependencies, which applies to eventNo and eventLoc fields

employee_contract

<u>eNo</u> PK FK	<u>contractNo</u> PK FK	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10

<u>eNo</u> PK	eName
1057	Hocine D
1068	White J
1135	Smith J

<u>contractNo</u> PK	eventNo	eventLoc
C1024	H25	Queens
C1025	H4	Yonkers
C1026	H25	Queens

<u>eventNo</u> PK	eventLoc
H4	Yonkers
H25	Queens

